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Spike Report: container-to-host inference reachability and datastore provisioning

Field	Value
Document ID	NOVA-SPK-003
Version	1.0
Status	Complete
Owner	Laxmi Poudel
Date	2026-09-09
Question	Can a container reach the host inference runtime on this platform, and does the datastore hold a 768-dimension vector with a working index?
Answer	Both, without configuration. Docker Desktop reaches a loopback-bound Ollama through its own host proxy, so the anticipated impediment does not occur here. `pgvector` 0.8.6 accepts `vector(768)`, builds an HNSW cosine index over it, and answers a nearest-neighbour query
Decisions affected	DL-030, R-09, G4, NOVA-SAD-001 Â§7

Why this ran

R-09 recorded container-to-host inference reachability as the most probable early impediment, on the reasoning that Ollama binds loopback by default and a container is not on the host's loopback. It was the last open entry condition for implementation, and it gates the deployment composition: if the path needed a host-side binding change, that change would have to appear in the deployment documentation, in the threat model, and in the first-run instructions for anyone cloning the repository.

The datastore half was folded into the same run because it answers a question of the same shape and against the same runtime. ADR-0005 fixed the vector dimension at 768 on retrieval evidence, but that figure had never been put through a column definition or an index build. A dimension that cannot be indexed is not a decision.

Environment

OS	Windows 11 Home 10.0.26200
Container runtime	Docker Desktop, Linux engine, server 29.7.2 (linux/amd64), Compose 5.5.1
Datastore image	`pgvector/pgvector:pg17`, digest `sha256:cf134a76â€¦f8e6f`, pgvector 0.8.6
Inference runtime	Ollama 0.33.3, host install, default binding `127.0.0.1:11434`

Method

One script, `artifacts/reachability_check.sh`, covering both questions and removing the container it starts on exit.

Question 1. A throwaway `alpine/curl` container issues `GET /api/tags` against `host.docker.internal:11434` and the HTTP status is recorded. The probe was run twice, once with Ollama on its default loopback binding and once with `OLLAMA_HOST=0.0.0.0:11434`, so that the default configuration is measured rather than assumed. Where `host.docker.internal` resolved to was recorded in both cases.

Question 2. A `pgvector/pgvector:pg17` container is started, then, in order: the extension is created and its version read; a table with a `vector(768)` column is created; an HNSW index with `vector_cosine_ops` is built over it; two rows are inserted, one a constant vector and one random, and a nearest-neighbour query is issued against the first; and the on-disk size of one vector is read with `pg_column_size`.

The second row exists so the ordering means something. A nearest-neighbour query against a single-row table returns that row whether or not the operator works.

Results

```
--- 1. container to host inference
  PASS  container reached host.docker.internal:11434 (HTTP 200)
```

```
--- 2. datastore with a 768-dimension vector
  PASS  pgvector 0.8.6 available
  PASS  vector(768) column accepted
  PASS  HNSW cosine index built over 768 dimensions
  PASS  nearest-neighbour query returns the expected row
  storage: 3076 bytes per 768-dimension vector

5 passed, 0 failed
```

Five of five, with Ollama on its default binding. Raw log: `artifacts/run-log-g4-2026-09-09.txt`.

Configuration	Host listener	Container to `host.docker.internal:11434`
Ollama default	`127.0.0.1:11434`	HTTP 200
`OLLAMA_HOST=0.0.0.0:11434`	`0.0.0.0:11434`	HTTP 200

Findings

1. The anticipated impediment does not occur on Docker Desktop, and the reasoning behind it was wrong. Inside a container, `host.docker.internal` resolves to `fdc4:f303:9324::254`, an address belonging to Docker Desktop's own host proxy rather than to the host's network interfaces. The proxy terminates the connection and forwards it to the host, loopback included. The container is therefore never a peer on the host's network in the way R-09 assumed, and a loopback-bound service is reachable. It works without `--add-host=host.docker.internal:host-gateway`, which the script passes anyway for portability.

2. Ollama stays on loopback, which is the better security position. The mitigation R-09 anticipated was `OLLAMA_HOST=0.0.0.0` plus a firewall rule scoping the port. That binding exposes an unauthenticated API that can load models and run generation to every host the firewall permits, and a firewall rule is the only thing standing in front of it. Not needing it removes a control that would have had to be documented, justified in NOVA-TM-001, and correctly reproduced by anyone cloning the repository. The deployment view's claim that services carry no inbound surface beyond the host holds for the inference runtime as written.

3. The finding is specific to Docker Desktop, and this is the part that will bite someone else. A Linux-native Docker Engine has no host proxy. There, `host.docker.internal` is undefined unless declared, and `host-gateway` maps to the bridge address, which a loopback-bound service does not answer on. The remediation R-09 described remains correct for that case, so the script retains it as its failure message rather than dropping it. What was measured is this platform, not the general case.

4. 768 dimensions survives contact with the schema. The column type is accepted, the HNSW cosine index builds over it, and the ordering is correct. ADR-0005 can move from a retrieval argument to a migration.

5. A 768-dimension vector costs 3076 bytes. Four bytes per dimension plus a four-byte header, which is `pgvector`'s uncompressed `vector` type behaving exactly as documented. At the retention volumes in NOVA-SRS-001 the embedding column is not a capacity concern, and the arithmetic is now recorded rather than estimated.

6. The check found one defect, in itself. The nearest-neighbour insert used a correlated reference inside `array_agg`, which PostgreSQL rejects as an aggregate over the outer query. The first run failed there while the four checks around it passed. A verification script that has never run is not evidence, which is the argument for running it before the deployment work rather than alongside it.

Limitations

1. One platform, one run. Docker Desktop on Windows 11 with the Linux engine. Neither Docker Engine on Linux nor Podman was tested, and finding 3 is reasoning about them, not measurement.
2. The reachability probe is `GET /api/tags`. It proves the HTTP path, not that a generation request under load behaves, and says nothing about latency across the proxy.
3. Docker Desktop updated itself from Compose 5.4.0 to 5.5.1 during the session, which is a reminder that the runtime version under this finding moves without being asked.
4. The datastore ran with default configuration and a throwaway container, no named volume, no tuning, no restart. Persistence across container replacement is the deployment composition's problem and is not evidence from this run.
5. The index was built over two rows. That exercises the type and the operator, not recall or build time at any realistic row count.
6. `pg_column_size` reports the stored attribute, not the index's own footprint.

Follow-up

#	Action	Why
1	Write the deployment composition against a loopback-bound inference runtime	The default configuration is now the verified one
2	Record the Linux-engine caveat in the deployment documentation	Finding 3. The first person to clone this on Linux hits it, and the script's failure message is the only place it currently lives
3	Pin the datastore image by digest in the composition	The digest is recorded here; drift is silent otherwise
4	Re-measure HNSW build time and recall at a realistic memory count	Finding limitation 5. Belongs with the M3 memory work, not here

Reproducing it

Requires a running container runtime and Ollama on the host. Nothing is left running.

```bash
bash docs/spikes/artifacts/reachability_check.sh
```

`PG_IMAGE` and `OLLAMA_PORT` are overridable. The script exits non-zero if any check fails.

Revision history

Version	Date	Author	Change
1.0	2026-09-09	Laxmi Poudel	Initial report. G4 closed, R-09 closed